AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 7 · STUDENT LESSON

Iterative Prompting

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Diagnose the failure. Change one component. Keep the record.

Revise prompts through paired evidence while preserving unresolved cases and unknown outcomes.

Keep one habit: Comparison contract → complete P/Q outputs → three-version log → separate quantitative review → independent design.

MAIN LESSON · SECTION 1

Which recorded failure should the next revision address?

> Change one prompt component and preserve the comparison evidence.

Your learning goals

I can explain the **mechanism** and terminology of controlled prompt experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall the rubric and evidence boundary

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Examples and Rubrics: keep the approved factual source, observable criteria and mandatory gates separate from a format example. Use the same rubric when comparing a prompt revision.

Your end goal

By the end, I can revise one prompt component and compare paired **output**s using the same **rubric**.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Examples and **Rubric**s. Give an example and a mistake that the idea helps you avoid.

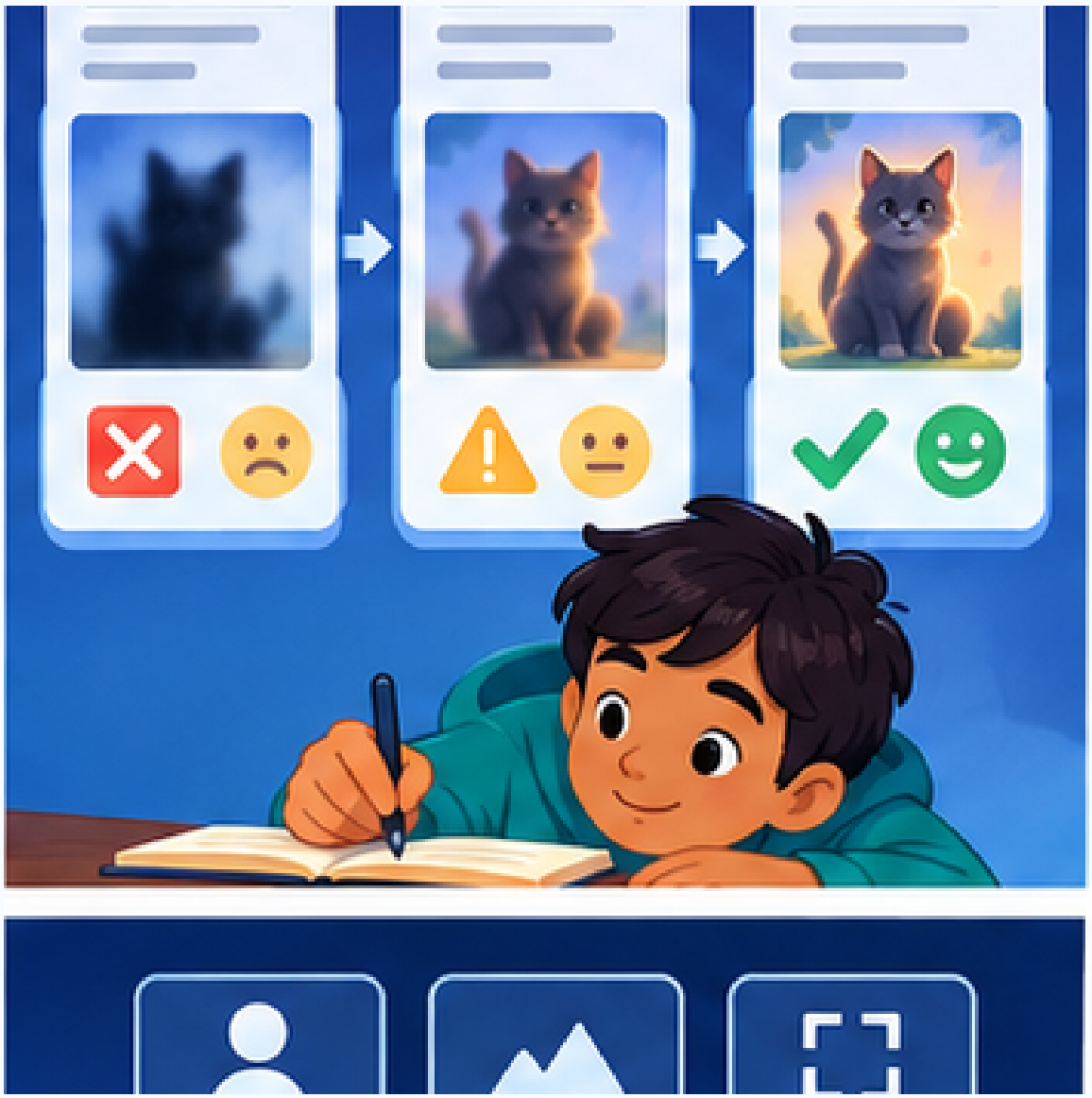
Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 7: Iterative Prompting

Revision should respond to diagnosed **output** failures.



CREATOR MISSION

Keep a three-**version** prompt log with reasons for each change.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Diagnose before editing the prompt

> The task, source, settings and rubric belong in the experiment log.

The controlled-revision idea

A prompt experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Label constructed results and missing evidence

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

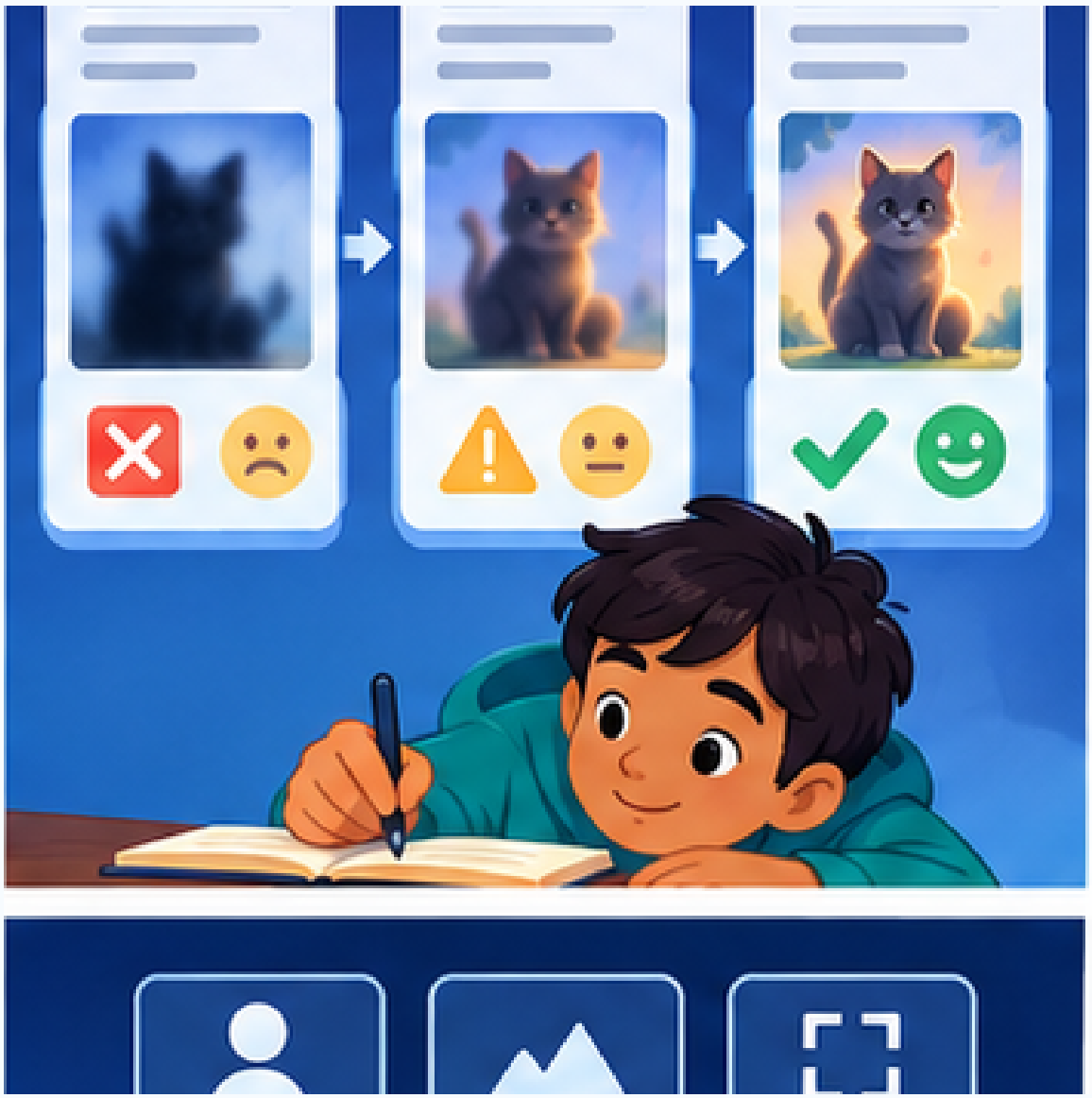
These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use iterative prompting to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Compare paired cases without hiding a regression

> The six-case W2 record is a separate quantitative review, not the four-question pictured experiment.

Freeze cases, sources and evaluation conditions

Evaluate both prompts on **paired case**s with fixed model/**version**, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

Score each case rather than counting attractive sentences.

Read the complete six-case record

Separate six-case development record	P	Q
Passing IDs	1,2,3,4	1,2,3,5,6
Missing-approval slice IDs4,6	passes4, fails6	fails4, passes6
All-case denominator	6	6

Keep denominators and the raw record

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain what the visual evidence represents

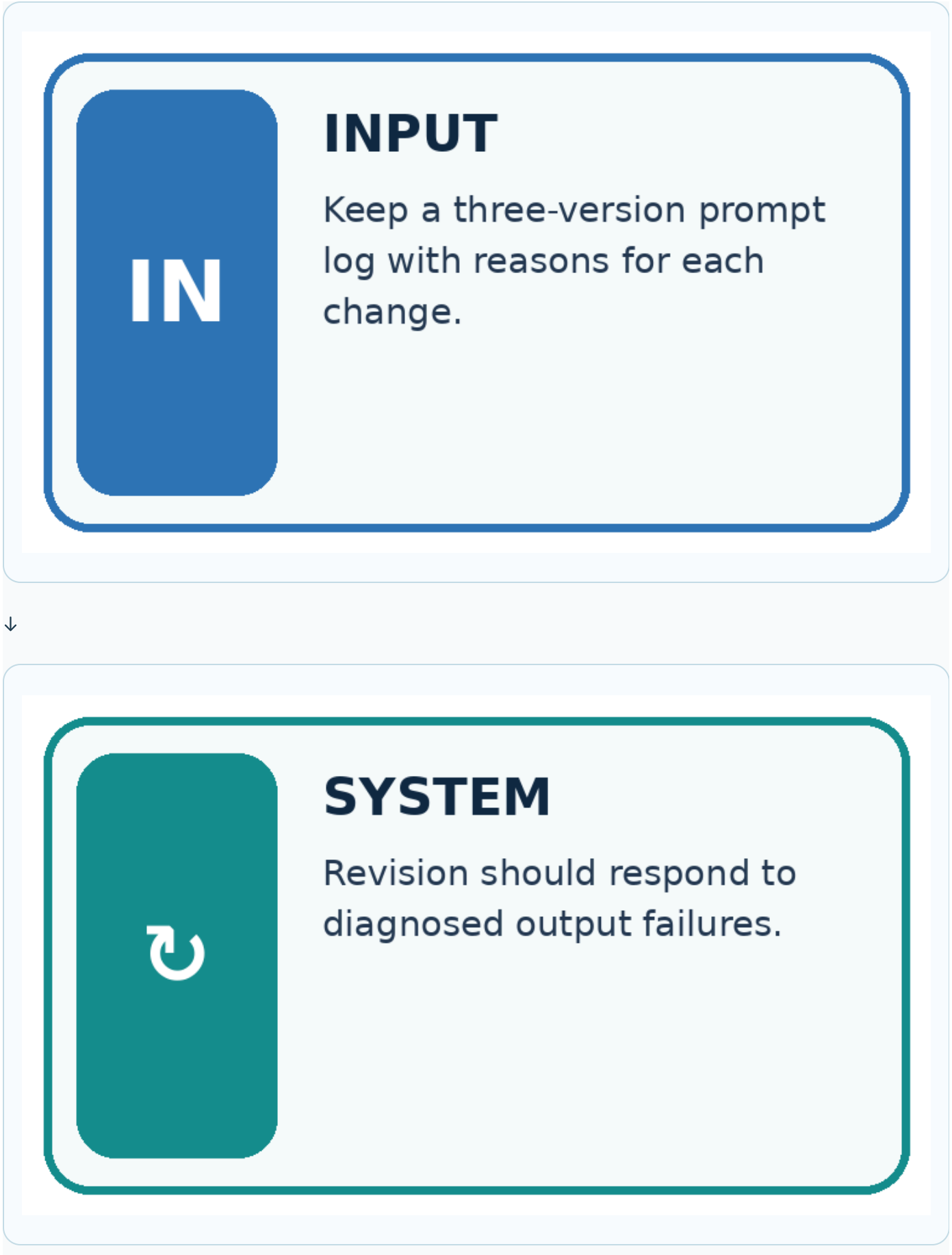
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Keep a three-version prompt log with reasons for each change.
↓
SYSTEM
Revision should respond to diagnosed output failures.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Controlled prompt experiments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.	Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric.	P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

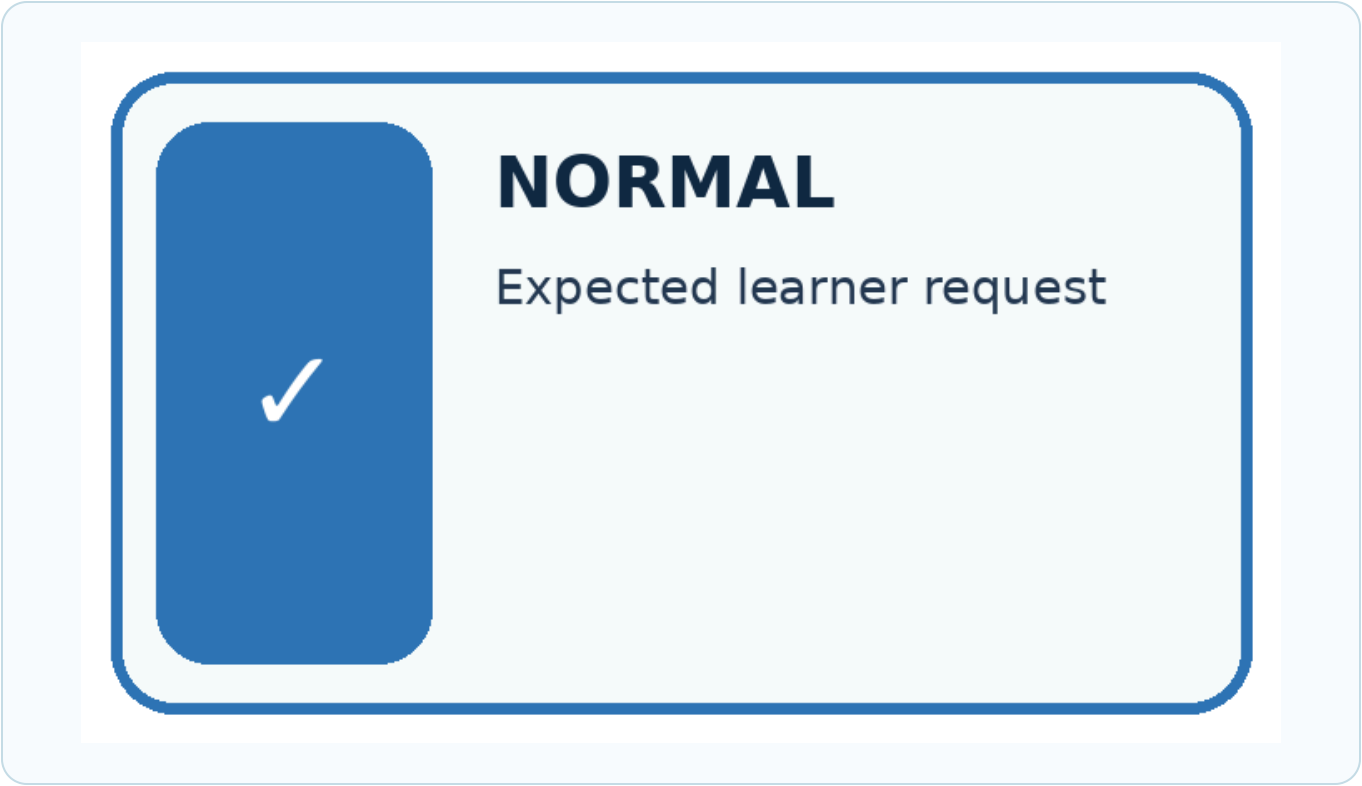
Connect the picture and the explanation

MAIN LESSON · SECTION 4

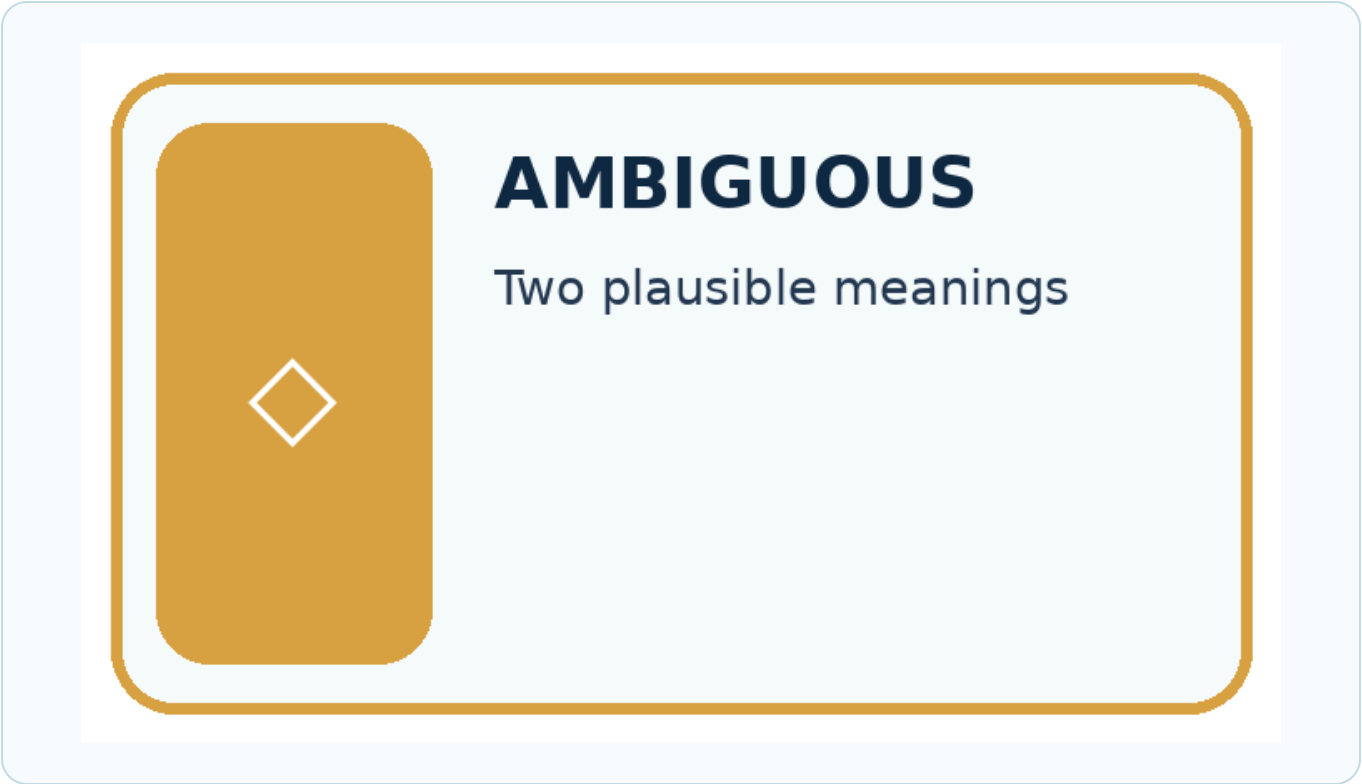
Compare a focused revision with simultaneous unrelated changes

> A better-looking output does not identify which change helped.

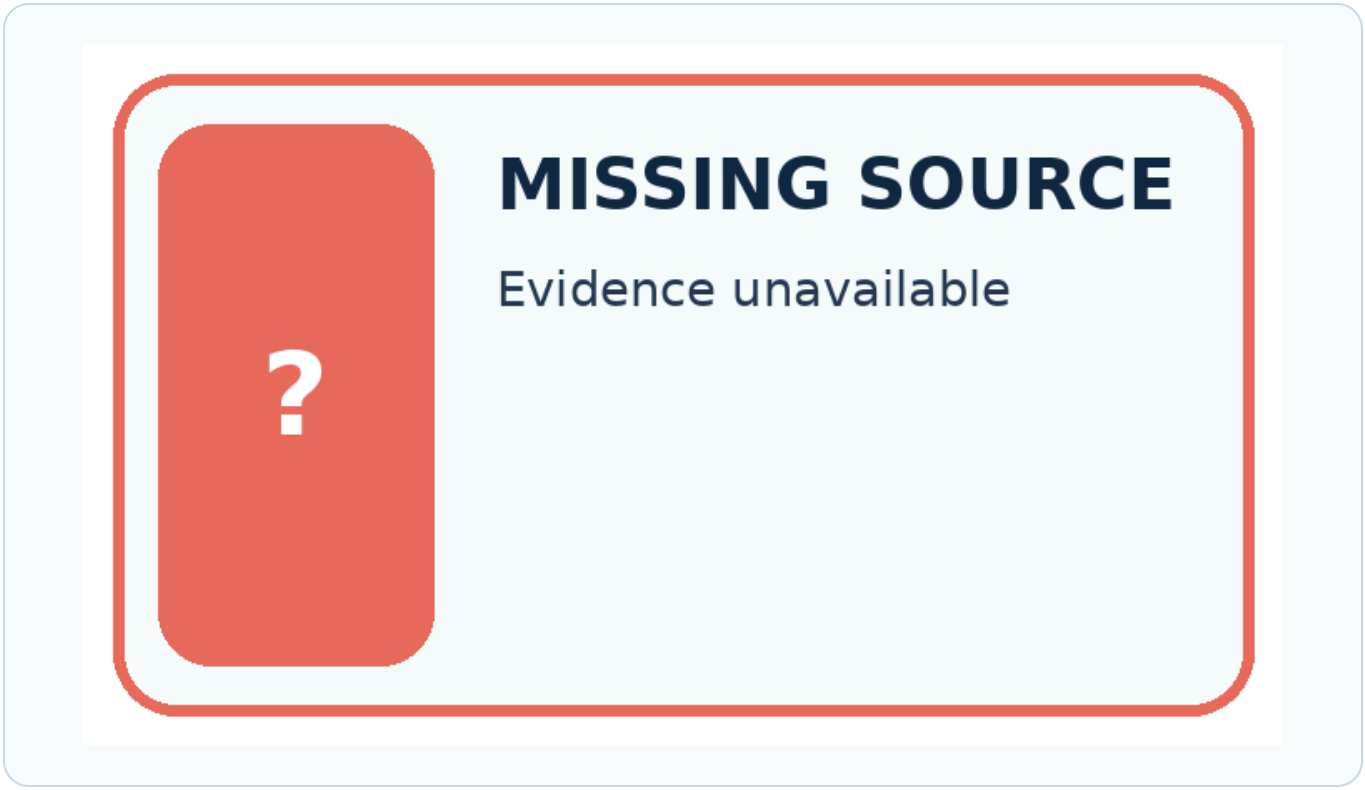
Four pictures with a purpose



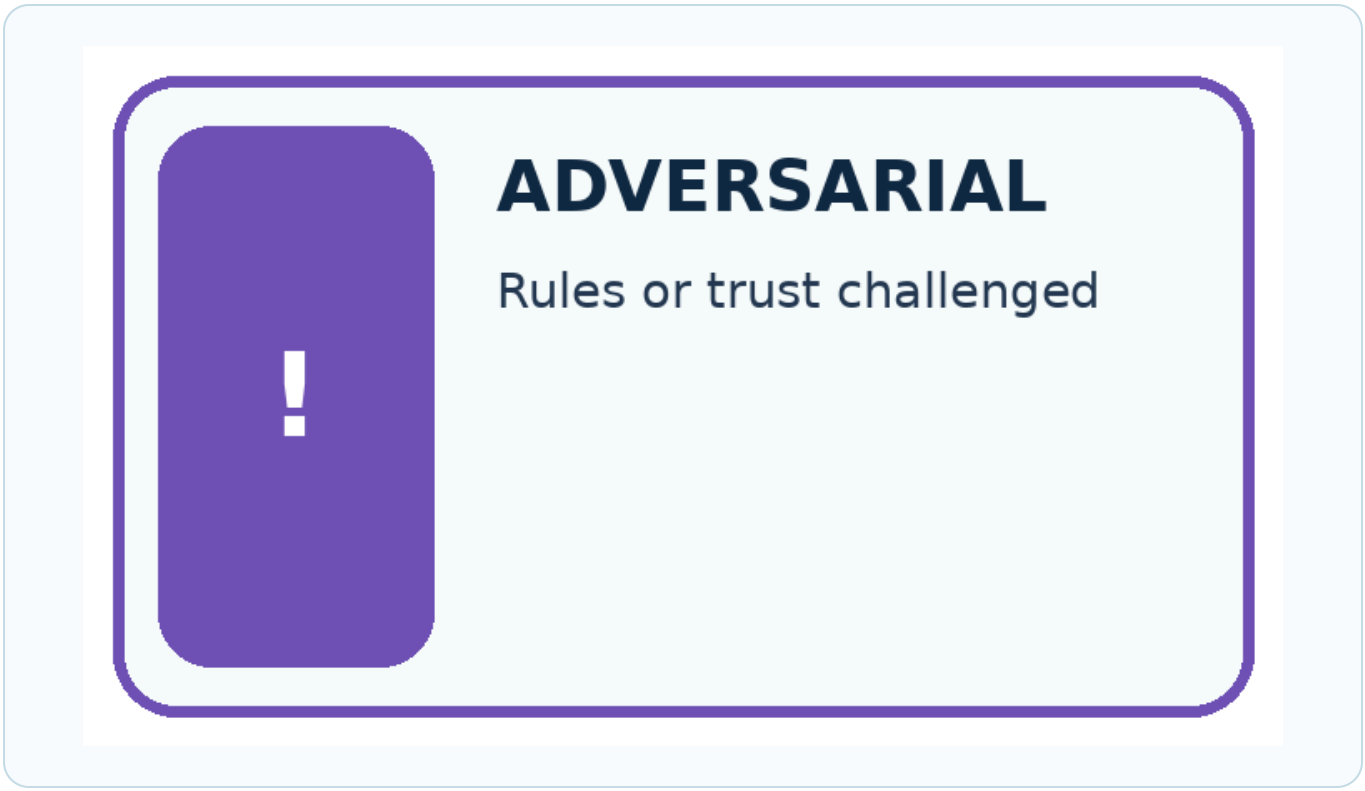
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Change model settings and note wording at the same time as the prompt.

▼ Check after predicting

Prediction: The comparison no longer isolates the prompt revision.

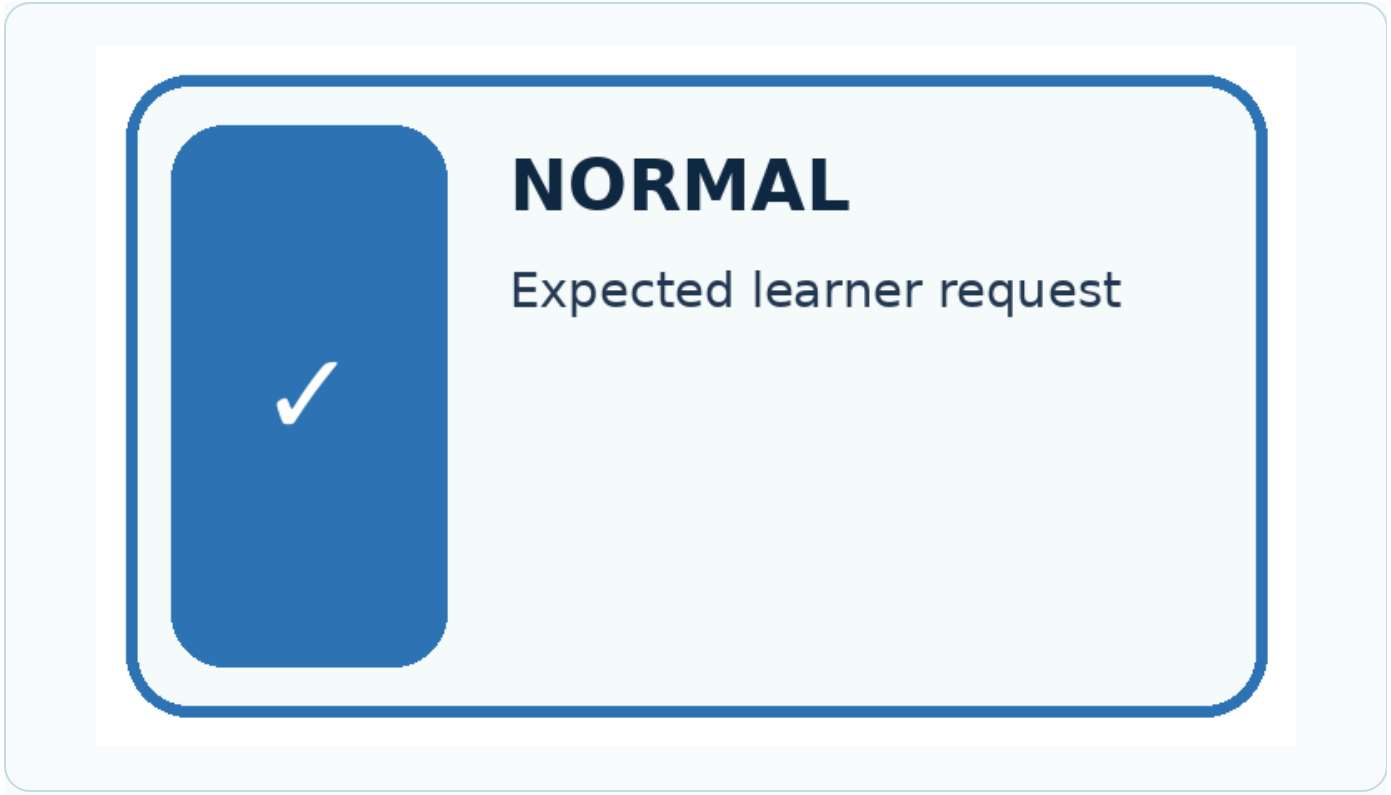
Reason: Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

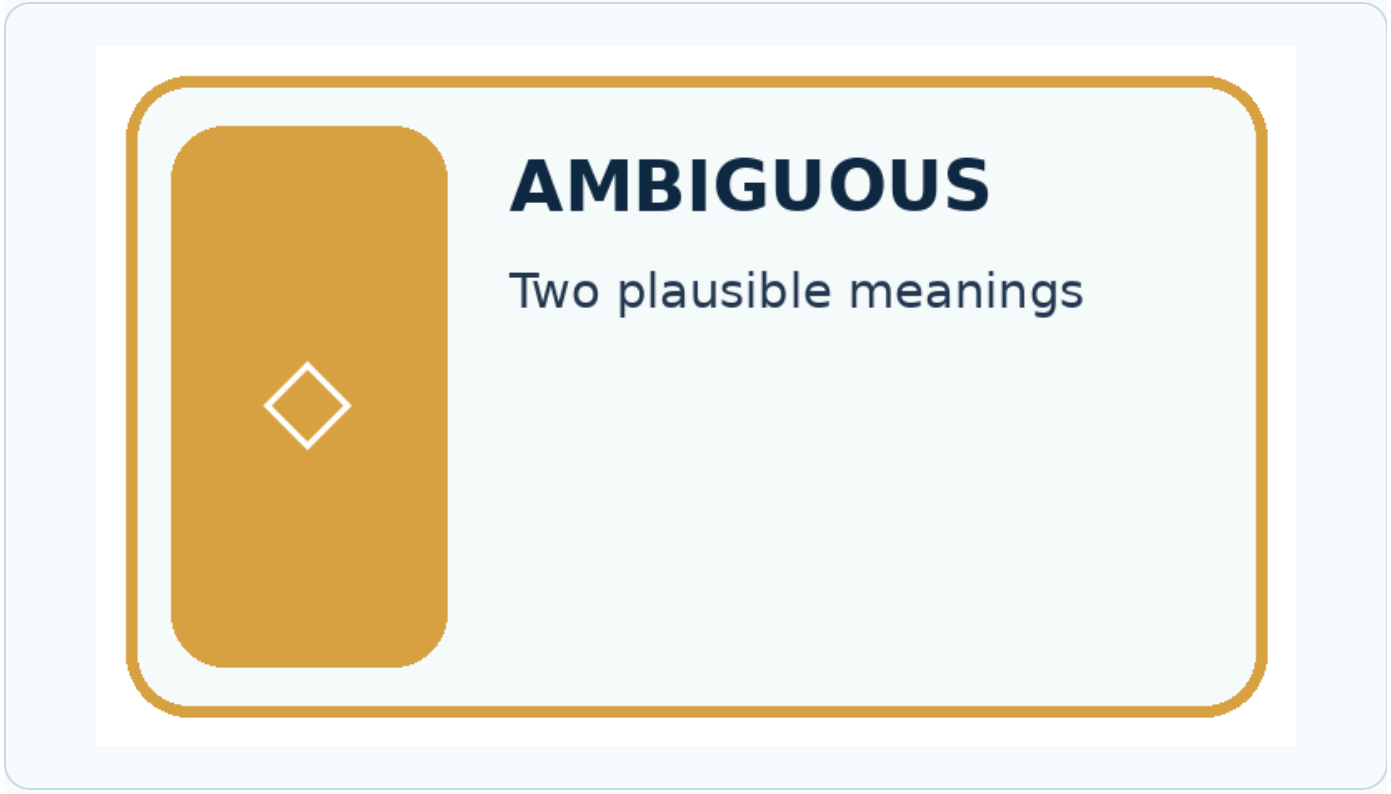
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace a separate four-question prompt revision

> A case-level gain leaves the unresolved failure visible.

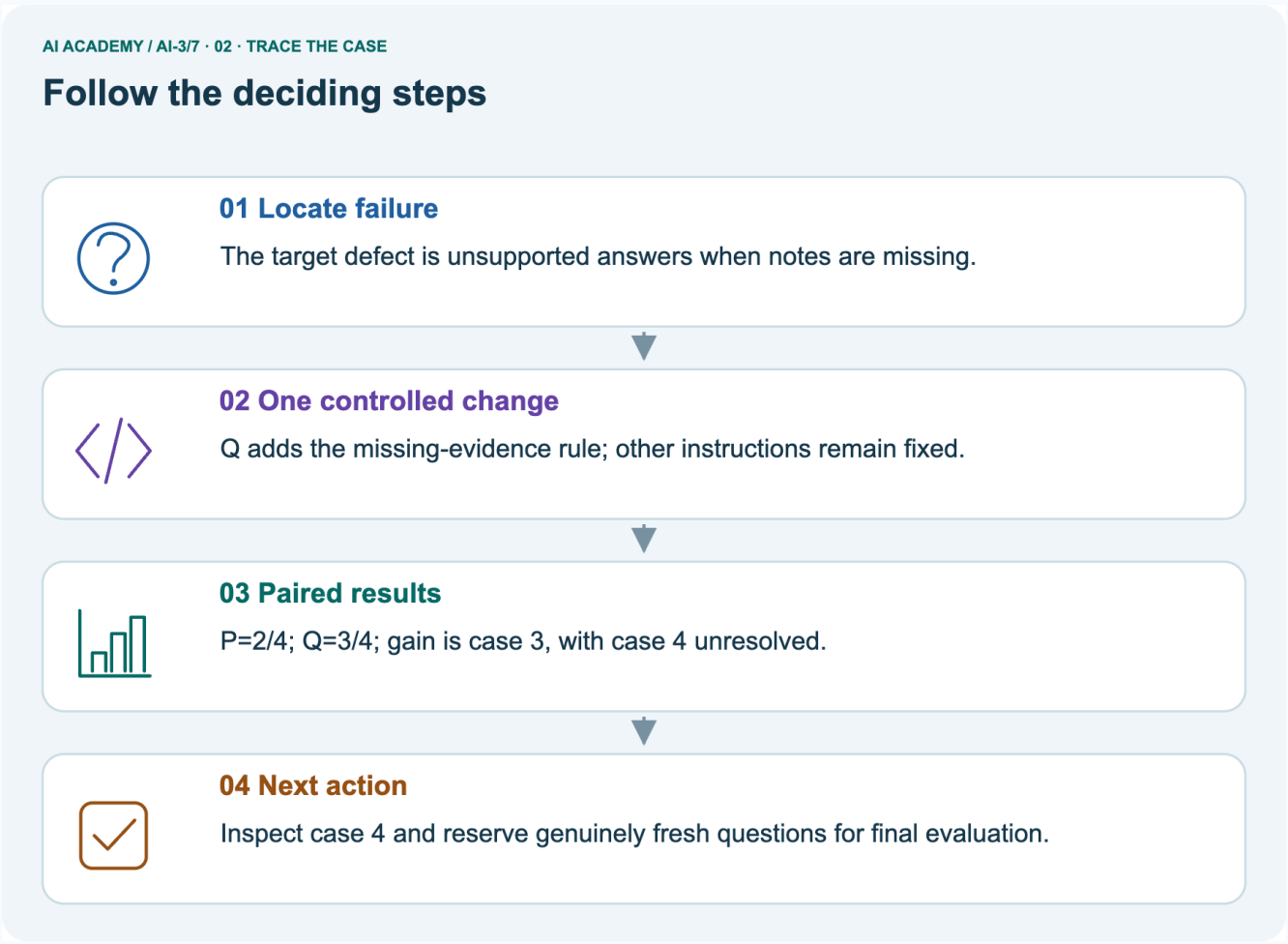
Read the pictured P-to-Q change and recorded outcomes

A fictional Study Buddy prompt invents a definition when approved notes omit it.

P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.

Trace the gain and remaining failure



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Locate failure	The target defect is unsupported answers when notes are missing.
2. One controlled change	Q adds the missing-evidence rule; other instructions remain fixed.
3. Paired results	P=2/4; Q=3/4; gain is case 3, with case 4 unresolved.
4. Next action	Inspect case 4 and reserve genuinely fresh questions for final evaluation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

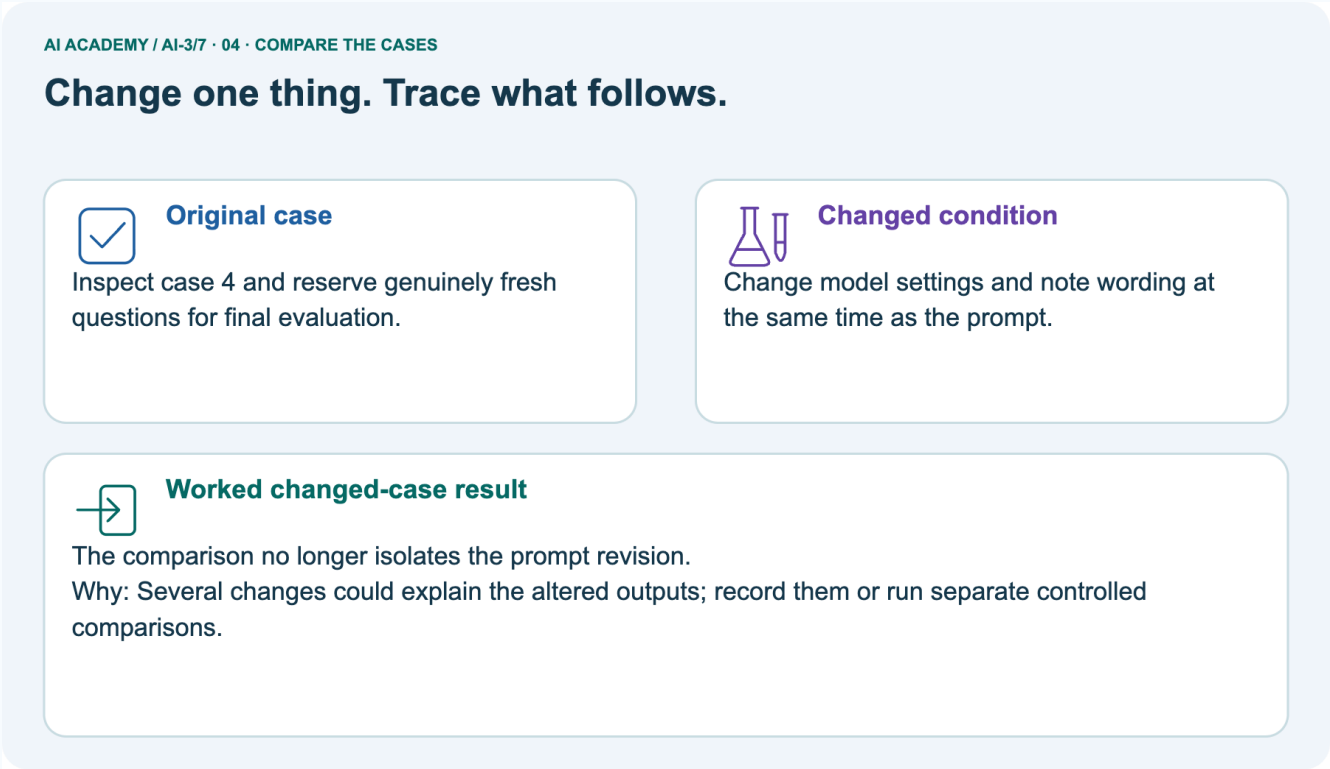
An **iteration** should respond to a recorded failure with a precise change. Comparing the same questions reveals whether other behavior changed too.

The invented results show a development gain of one case, not a guarantee that the new instruction always works. Save prompt text, settings, source notes and case-level outcomes.

Predict changing settings and sources too

- Change one thing: Change model settings and note wording at the same time as the prompt. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State what the small comparison cannot establish

- Stochastic **output** can vary across runs; a small comparison does not prove an average effect or classroom learning benefit.
- Read the labelled picture: What **evidence** supports improvement here, and what remains unproven?

▼ Review · The separate six-case W2 result

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

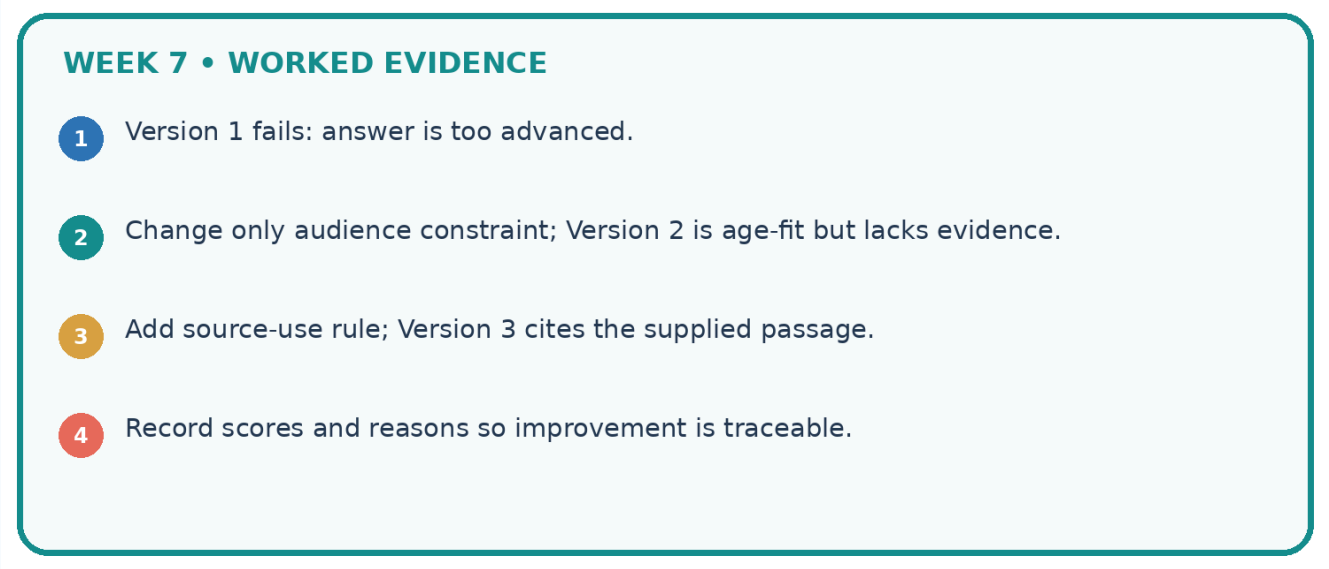
P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Version** 1 fails: answer is too advanced.

2. Change only audience constraint; **Version** 2 is age-fit but lacks **evidence**.

3. Add source-use rule; **Version** 3 cites the supplied passage.

4. Record scores and reasons so improvement is traceable.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

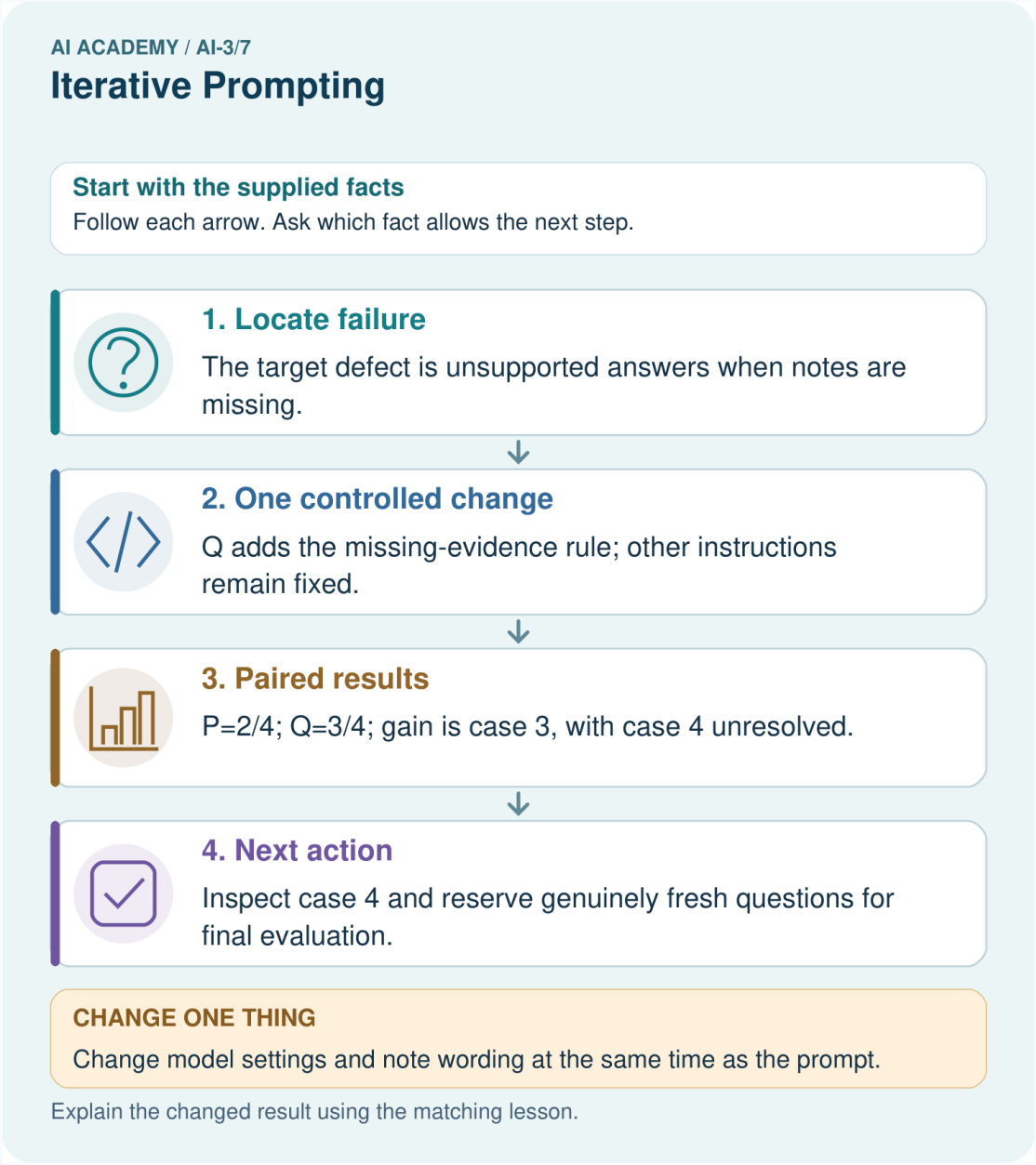
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Iterative Prompting

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Keep the remaining failure beside the gain

> An iteration record explains what changed and what is still unknown.

Repair the always-works argument

Claim to repair: Q passed one more question, so the missing-evidence instruction now always works. The supplied four-question record improves2/4 to3/4 but still fails case4. Inspect that output, preserve the failure and plan the next single revision. These are invented single-run development outcomes, not an estimate of average effect, untouched final performance or classroom learning benefit.

Propose a revealing next check

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Keep adding instructions until an output looks better. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise a paired-result trace with less help

> Use the separate six-case data for W2 and the four-question picture for its own discussion.

Protect final evidence after development selection

Prompt selection on development cases cannot be reported as independent final performance.

Complete the six-case W2 trace

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Controlled prompt experiments”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an all-future-cases claim

- Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the pictured improvement and limit

- What **evidence** supports improvement here, and what remains unproven?

Explain the picture

My explanation using a pictured step and the given facts

Keep independent work separate

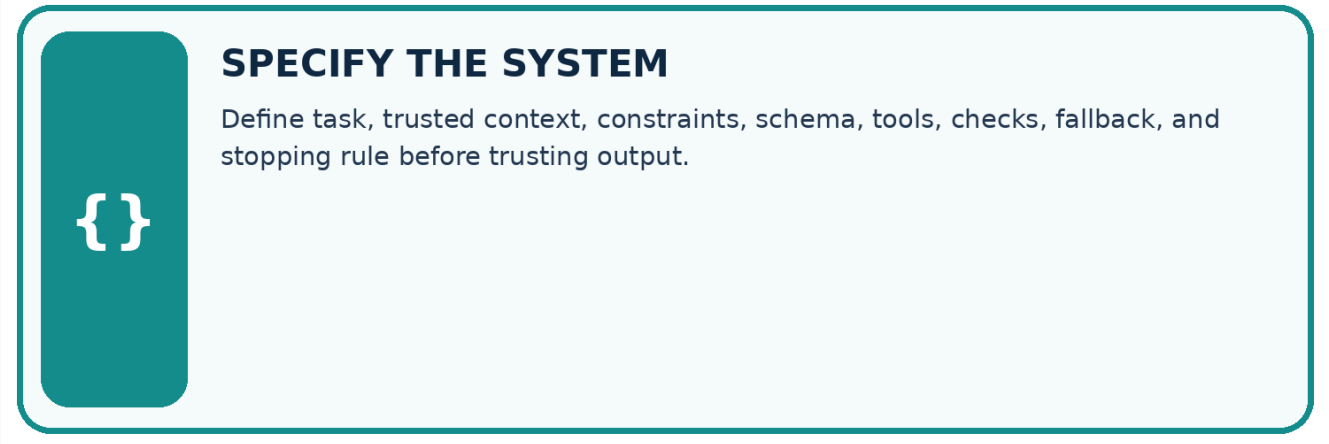
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Inspect complete outputs and keep a truthful three-version log

> The third revision is a proposal, not a fabricated measured improvement.

The approved source and fixed rubric

Complete fictional source N: condensation changes water vapour into liquid water; freezing changes liquid water into ice. No definition of melting or evaporation is supplied. Fixed task wrapper: for each requested term, produce exactly Term and Meaning using approved N. Full pass requires both fields, correct requested term and meaning supported by N; when no definition exists, the rubric expects not supplied.

Model/settings are declared fixed for these invented output records; no actual service/version is supplied or run.

Three complete prompt versions and evidence status

Version	Complete variable prompt text appended to the fixed wrapper	Status
P	Answer helpfully.	Supplied invented outputs below
Q	Answer helpfully. If approved notes lack the answer, say not supplied.	Supplied invented outputs below
R	Answer helpfully. If approved notes lack the answer, say not supplied. Before drafting, check whether N explicitly defines the requested term.	Proposed next version; not run

Four paired complete output cards

Case/request	Complete P field contents	Complete Q field contents
1 condensation	Term: condensation; Meaning: water vapour changes into liquid water	Term: condensation; Meaning: water vapour changes into liquid water
2 freezing	Term: freezing; Meaning: liquid water changes into ice	Term: freezing; Meaning: liquid water changes into ice

Case/request	Complete P field contents	Complete Q field contents
3 melting	Term: melting; Meaning: ice changes into liquid water	Term: melting; Meaning: not supplied
4 evaporation	Term: evaporation; Meaning: liquid water changes into vapour	Term: evaporation; Meaning: liquid water changes into vapour

Score, diagnose and plan the next controlled revision

1. Three-version log investigation: score every supplied P/Q output against N and the fixed rubric before seeing the pictured totals. Case3 improves because Q uses the required fallback; case4 remains unsupported even if its statement could be true outside this source-only contract. Record exact P/Q text, diagnosed defect, one added missing-evidence rule, per-case judgments and the unresolved case.

   For R, record the one added source-check step and predicted behavior before a future test; retain not run instead of inventing results. Plan identical development cases/settings and a prespecified repeated-run scheme if outputs vary. Fresh final cases must be reserved after selection. A partner can audit these paper judgments, but the invented records do not reproduce a real prompt experiment.

   Keep the three-bullet format-revision task for independent work.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

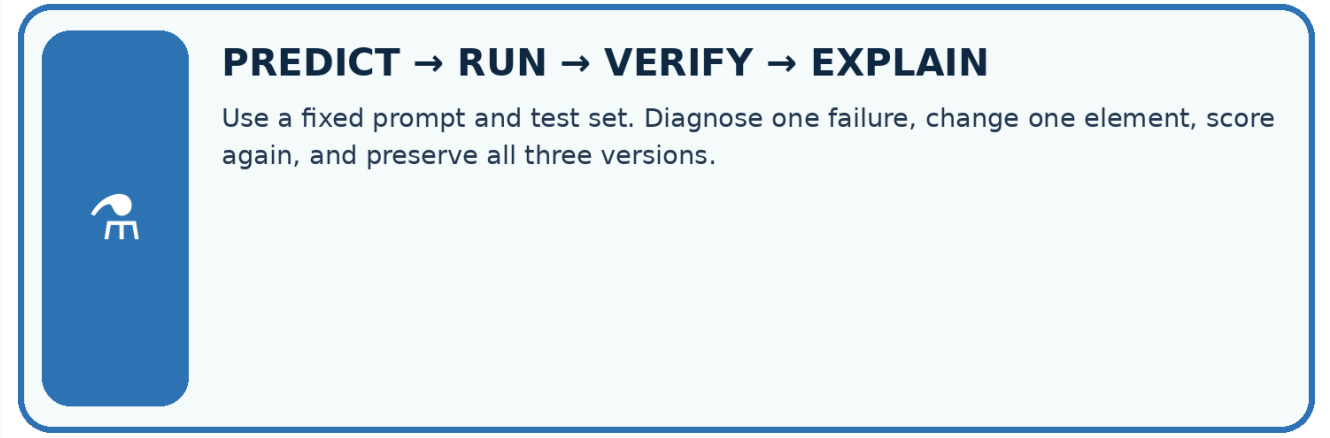
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the complete source N, fixed task/rubric, P/Q output cards and three-**version** prompt log. Score every supplied result, preserve case4, and label R as a proposed unrun revision. No actual model calls or measured third-**version** improvement are supplied.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: plan the fresh three-bullet format repair

> A correct format revision still needs a factual-preservation check.

Independent brief and scope

A factual response uses four paragraphs when the contract requires three bullets. Design a controlled prompt revision and evaluation.

Design the comparison; no revised outputs, source text or model scores are supplied. Fix the approved evidence, factual criteria, task cases and model/settings while changing the format instruction. Define three-bullet compliance separately from factual support, and preserve

unrun outcomes as unknown.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

Week 7 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You change a prompt's audience, length and source rules together; the result improves. What can you conclude, and how would you investigate the cause?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Maintain prompt **version**, diagnosed failure, change, and before/after **rubric** scores.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A summary has accurate facts but uses unexplained terms. Propose one prompt change and a comparison that checks both accessibility and factual preservation.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Link each project revision to a diagnosed failure

> Preserve the actual record and the boundary of its evidence.

This week's concrete contribution

Keep the Verified Study Buddy log as version, exact prompt, frozen task/source/settings, diagnosed criterion failure, one revision, predicted outcome, observed result or not-run status, regressions and remaining limits. Reuse the same development cases for controlled diagnosis; after selection, collect fresh final evidence.

Separate format compliance, factual support and useful explanation rather than promoting a favourite answer.

Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results.

Keep the checkpoint and human choice

THIS WEEK'S CHECKPOINT Maintain prompt **version**, diagnosed failure, change, and before/after **rubric** scores.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

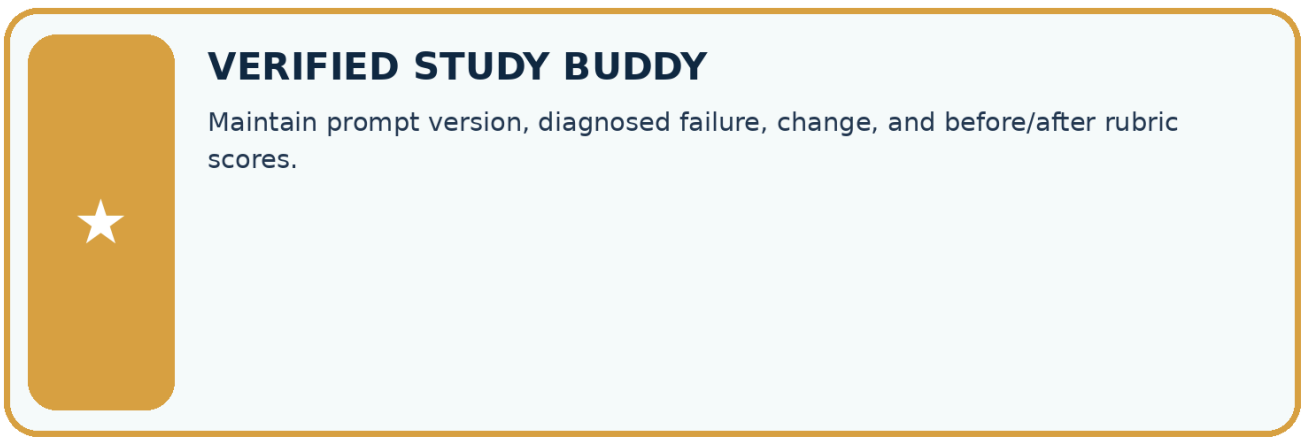
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the failure, one revision and comparison evidence

> Do not turn development selection into a final result.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of controlled prompt experiments

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
iteration	One repeat of a process, often followed by inspection and revision.	
failure diagnosis	An investigation locating where and why a process failed.	
version	An identified state of a document, dataset, model or program.	
change log	A record of what changed, when it changed and why.	

Terms for the connected explanation

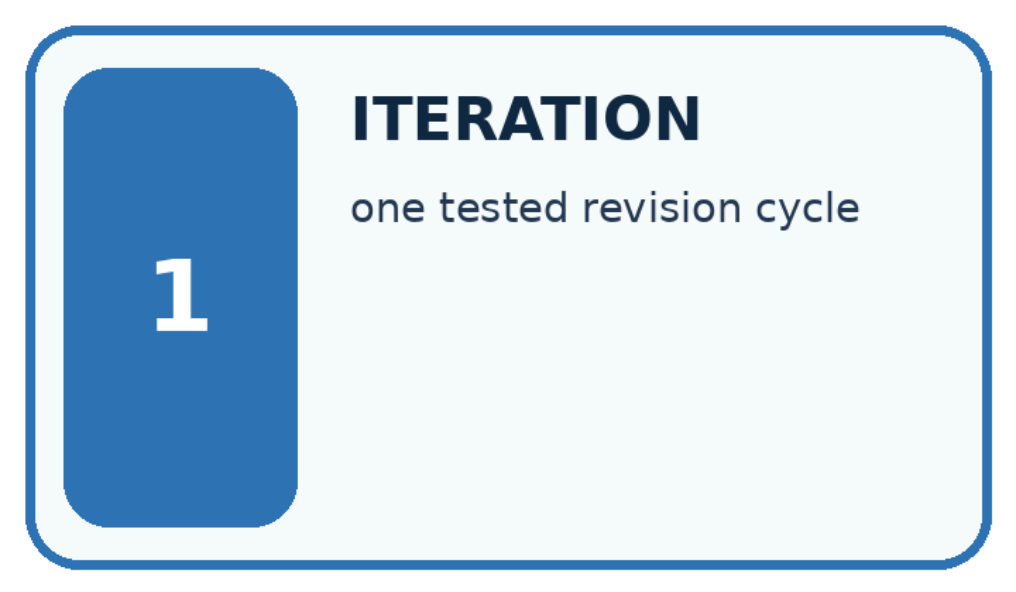
Word	Meaning
Paired case	The same case evaluated under each compared procedure
Rubric	Prespecified criteria for judging an output
Regression	A previously passing condition that fails after a change

▼ **Earlier wording, alternatives and picture transcripts**

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.









Words used in the closer explanation

▼ Choose a stronger thinking challenge

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

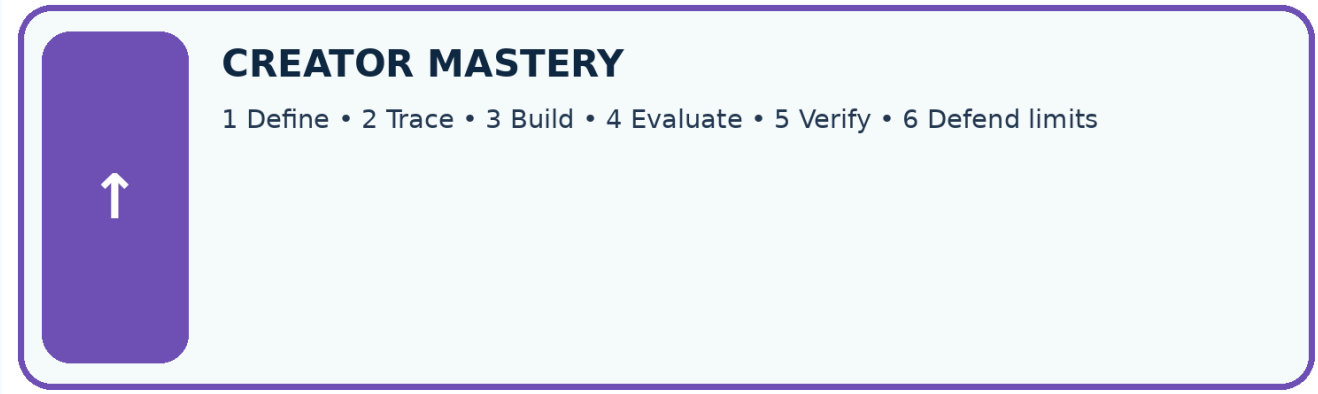
> **Write or draw your reasoning and evidence.**

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Design a new format-revision evaluation independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check and record

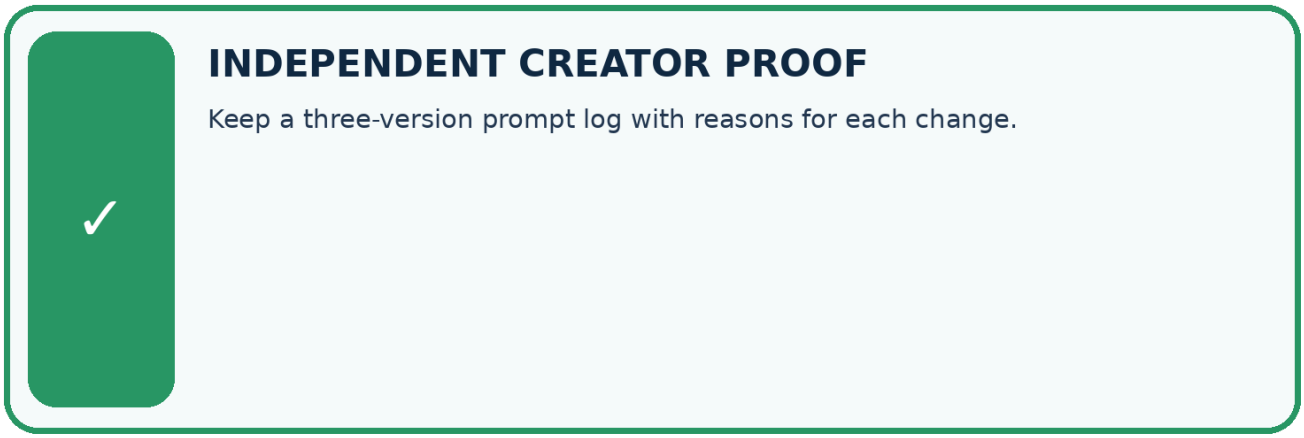
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer ‘why?’

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for explaining this week’s idea

Use after practice. Design a new format-revision evaluation independently is the selected independent case.



EXPLANATION CHECK Require one format change, fixed evidence and settings, separate factual checks and fresh final cases.

> **Write or draw your reasoning and evidence.**

▼ **Extra practice: Revise one cause and retest**

REFERENCE · SECTION 13

Extra practice: Revise one cause and retest

> Try the question before opening its worked explanation.

Read the supplied case and question

A factually correct explanation is too difficult for a nine-year-old.

Make and test one focused revision.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Constrain vocabulary and request a concrete example. Compare readability and factual preservation on the same passages.

Evidence or stage	Value or result
Diagnose	Too difficult for a nine-year-old
Revise	Simpler vocabulary and one concrete example
Retest	Use the same passages
Compare	Readability and factual preservation

▼ Earlier wording, alternatives and picture transcripts

Week 7 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Revise one cause and retest

A factually correct explanation is too difficult for a nine-year-old.

Make and test one focused revision.

Constrain vocabulary and request a concrete example. Compare readability and factual preservation on the same passages.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of controlled prompt experiments	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional Study Buddy prompt invents a definition when approved notes omit it.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy prompt invents a definition when approved notes omit it.
- Which labelled step matches this detail? Inspect case 4 and reserve genuinely fresh questions for final evaluation.
- 1. Locate failure
- 2. Next action
- 3. One controlled change

- 4. Paired results

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Next action Inspect case 4 and reserve genuinely fresh questions for final evaluation. An **iteration** should respond to a recorded failure with a precise change.

Comparing the same questions reveals whether other behavior changed too. The invented results show a development gain of one case, not a guarantee that the new instruction always works.

Save prompt text, settings, source notes and case-level outcomes.

Changed-case question

- Changed-case practice: Change model settings and note wording at the same time as the prompt. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. The comparison no longer isolates the prompt revision.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The comparison no longer isolates the prompt revision. Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Repair and save your work

Review Examples and **Rubric**s (ai-3/6). Goal: Distinguish a format example from approved factual evidence. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card.. Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?.

Return to this week and explain what you changed.

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

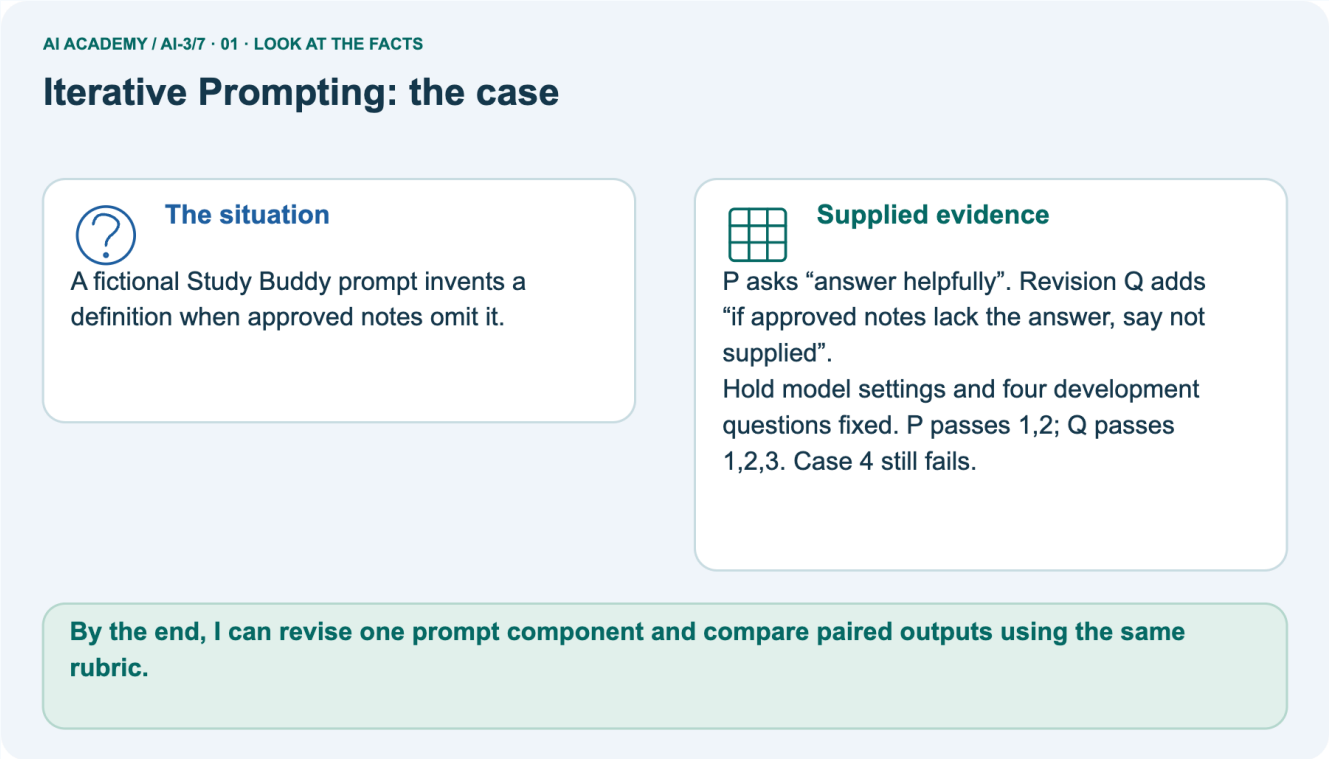
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Iterative Prompting: the case



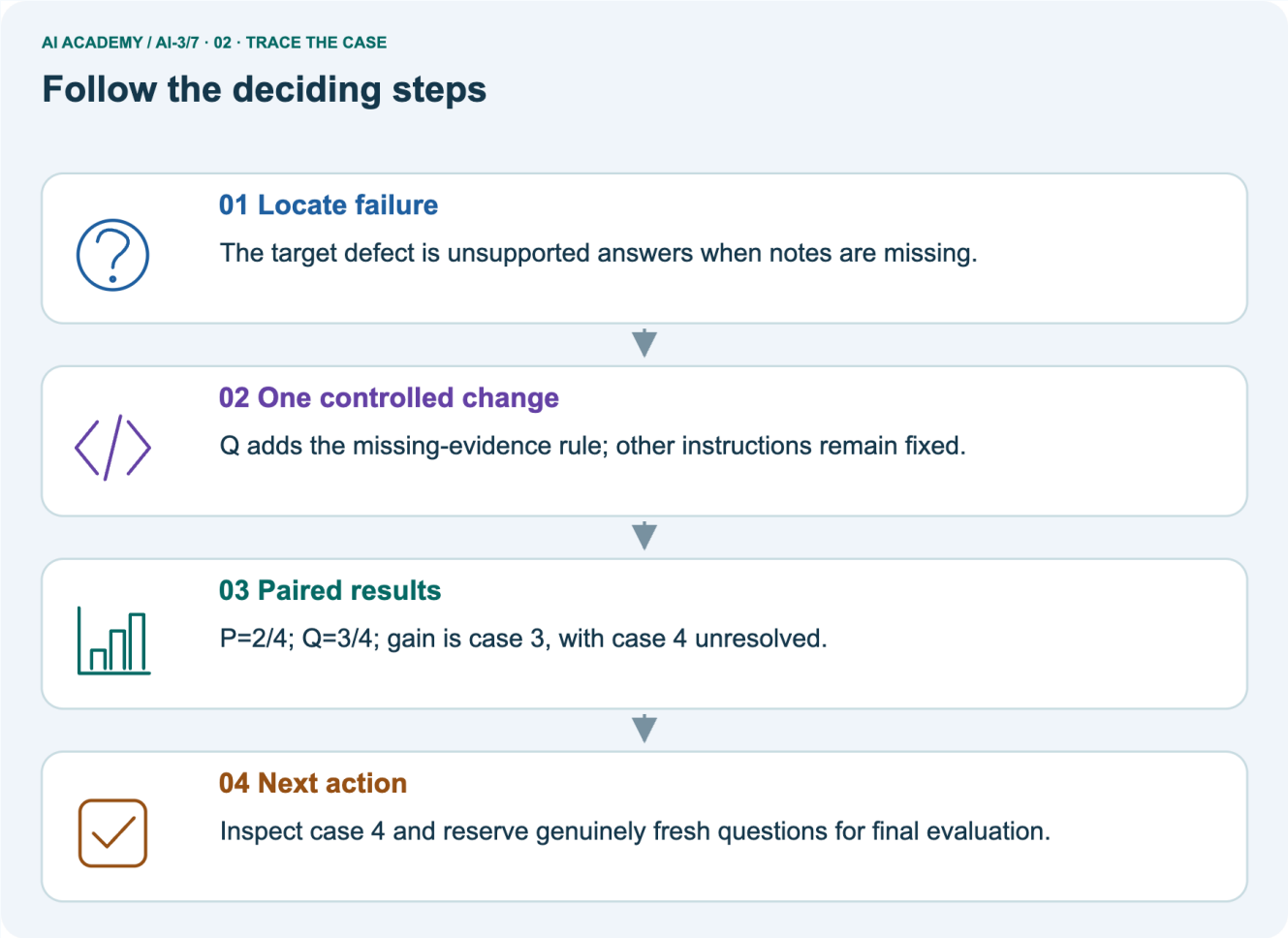
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy prompt invents a definition when approved notes omit it. Supplied facts: P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

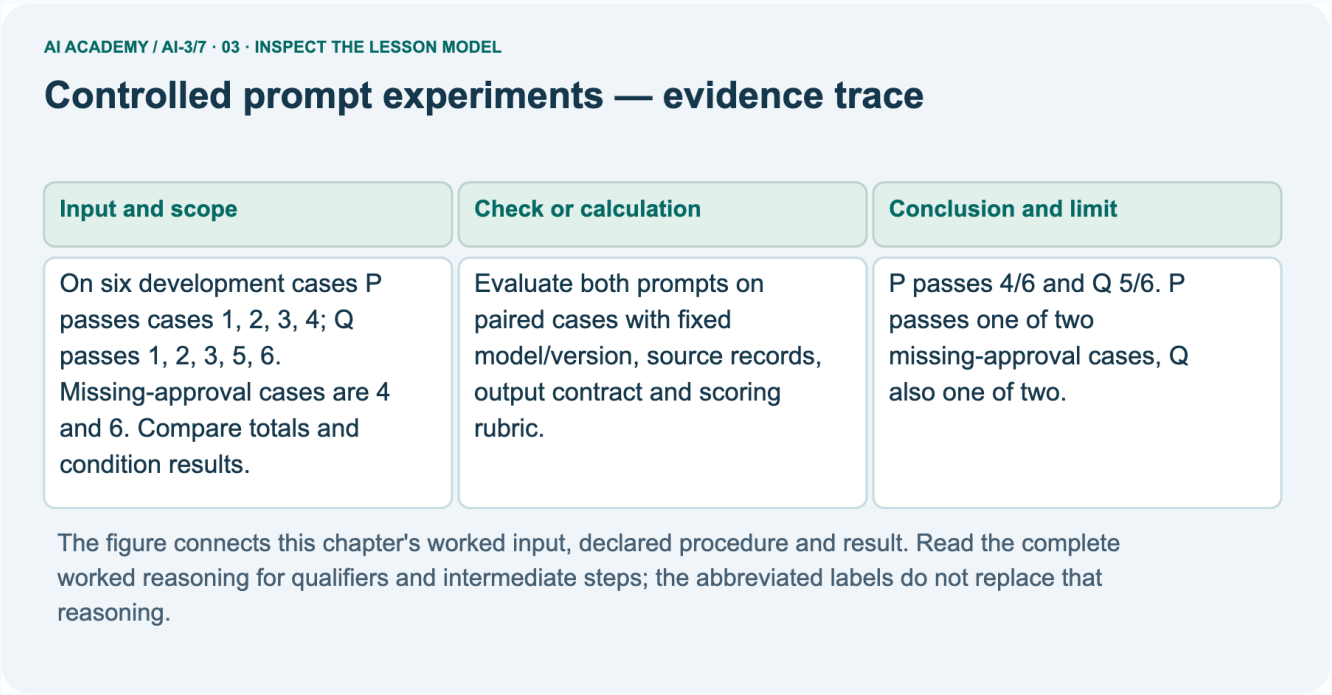


Follow the labelled picture with the matching explanation. Open the vector diagram

Locate failure: The target defect is unsupported answers when notes are missing. → One controlled change: Q adds the missing-**evidence** rule; other instructions remain fixed. → Paired results: P=2/4; Q=3/4; gain is case 3, with case 4 unresolved. → Next action: Inspect case 4 and reserve genuinely fresh questions for final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

03. Controlled prompt experiments — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

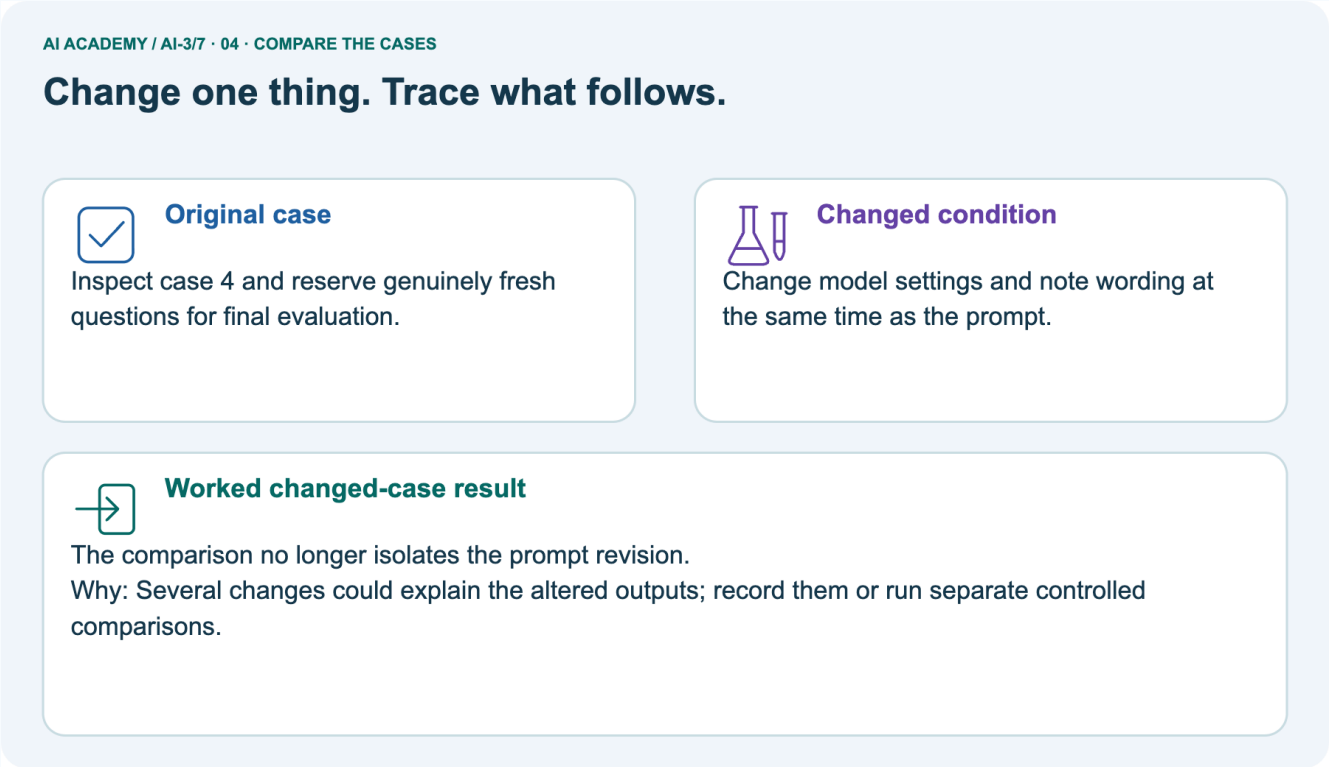
Controlled prompt experiments — **evidence** trace. On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6.

Compare totals and condition results. | Evaluate both prompts on **paired case**s with fixed model/**version**, source records, **output** contract and scoring **rubric**. | P passes 4/6 and Q 5/6.

P passes one of two missing-approval cases, Q also one of two.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



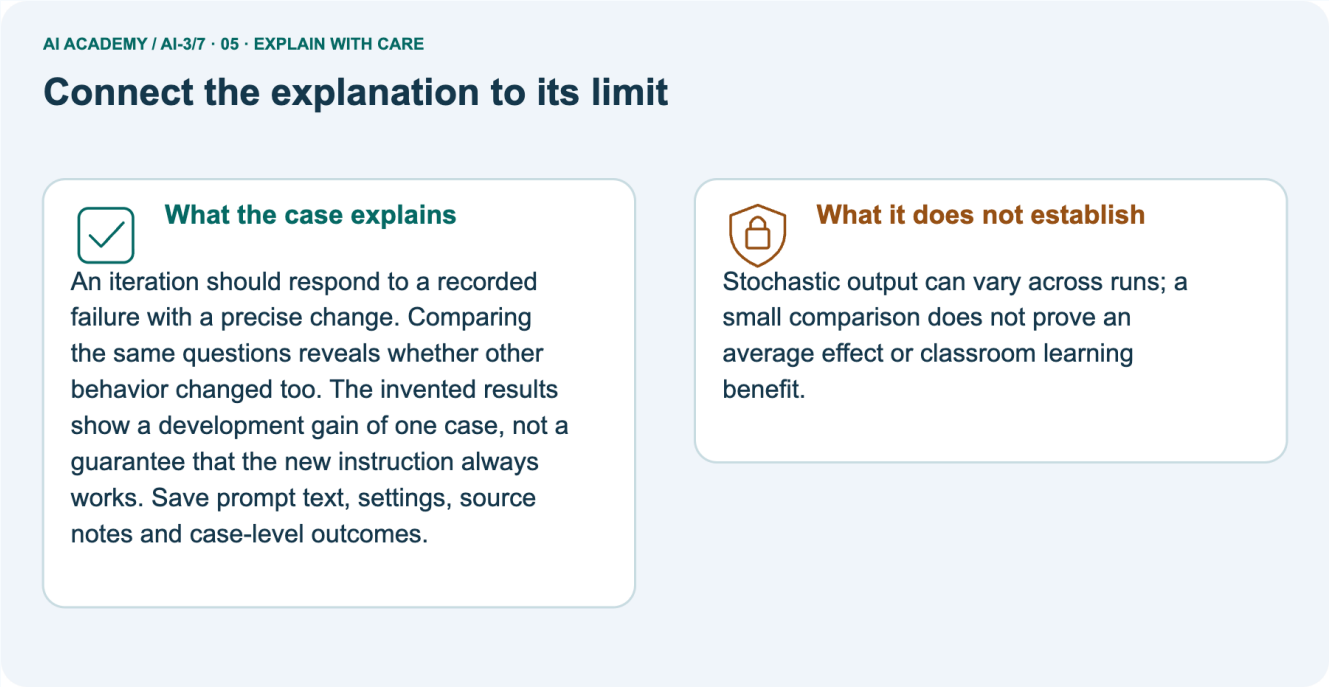
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect case 4 and reserve genuinely fresh questions for final evaluation. Changed condition: Change model settings and note wording at the same time as the prompt.

Expected result: The comparison no longer isolates the prompt revision. Reason: Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

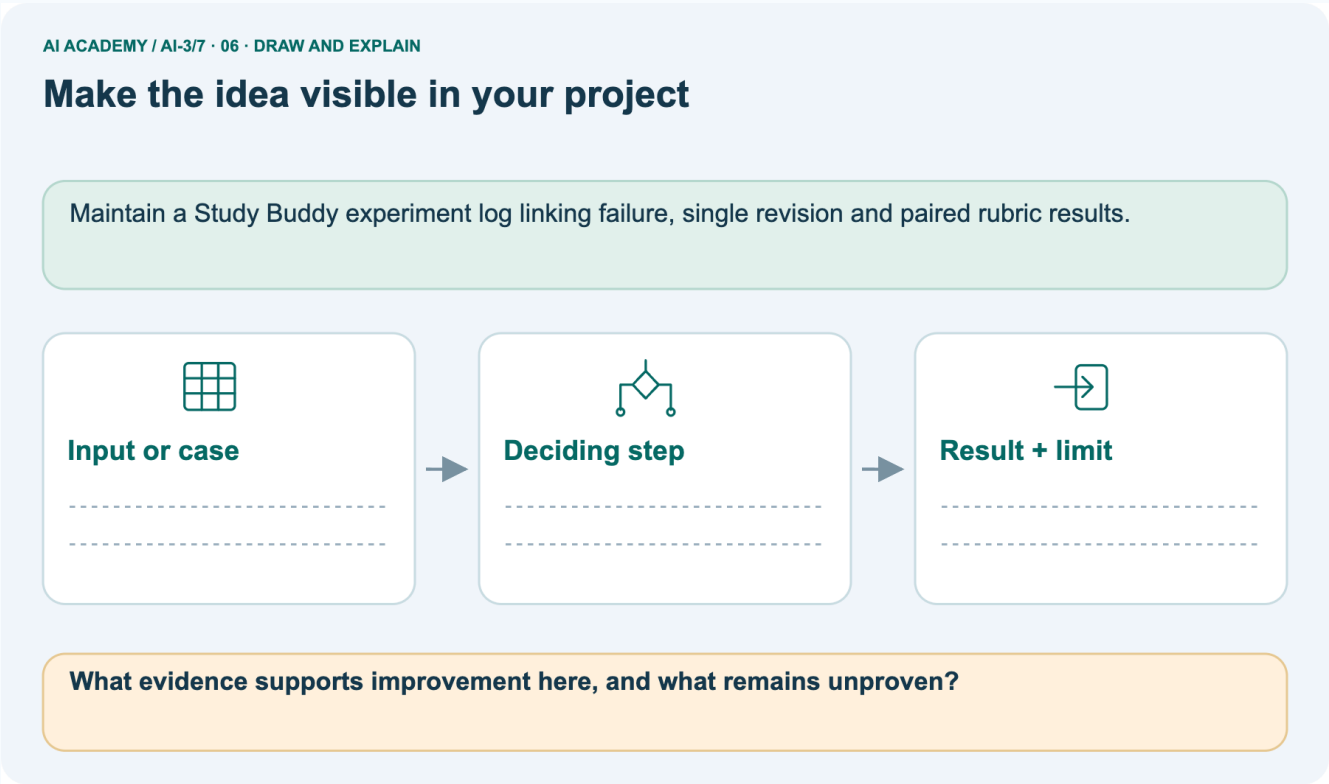
An **iteration** should respond to a recorded failure with a precise change. Comparing the same questions reveals whether other behavior changed too.

The invented results show a development gain of one case, not a guarantee that the new instruction always works. Save prompt text, settings, source notes and case-level outcomes.

Limit: Stochastic **output** can vary across runs; a small comparison does not prove an average effect or classroom learning benefit.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** supports improvement here, and what remains unproven?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Iterative Prompting: the case

02. Follow the deciding steps

03. Controlled prompt experiments — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.